

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

Question Paper No. : UGF042

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. The amount of thrust upon water to propel a boat is given by $f(\theta) = \frac{\theta - \sin \theta}{2\pi\theta^3}$. What thrust limit you can expect if the Pitch angle θ is turned towards the zero angle.
2. If $\sinh y = -\frac{3}{4}$ then identify the value of y .
3. Determine the general value of $\log(1+i)$.
4. Let A be a square matrix of order two. If $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are latent vectors corresponding to the latent values 1 and 5 of A respectively. Find the elements of A .
5. The current age distribution vector and age transition matrix of rats are respectively $\bar{x}_1 = \begin{bmatrix} 24 \\ 24 \\ 20 \end{bmatrix}$ and $A = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}$. After 1 year what will be the rats' age distribution vector.

6. Construct a fourth-degree equation whose roots are $1+i$ and $\sqrt{2}$.
7. If α, β , and δ are the roots of $x^3 - 7x^2 + 6x - 4 = 0$ then determine $\sum(\alpha\beta)^2$.
8. Calculate the value of the integral $\int_0^{\infty} \frac{x^8(1-x^6)}{(1+x)^{24}} dx$.
9. Determine the volume of the bounded region $R: 0 \leq x \leq 1, 1 \leq y \leq 2$, and $2 \leq z \leq 3$.
10. Determine the area of the region bounded by, $x = 0, x = 1; y = 0, y = x$.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. (a) In a celestial navigation, while measuring the position of the object an error occurred in replacing $\frac{1}{6}(8\sin\theta - \sin 2\theta)$ by θ . Assuming θ small, calculate the approximate error in the measurement. (7)
- (b) Determine $\cos 2\theta \cosh 2\phi$ when $\sin(\theta + i\phi) = \tan \beta + i \sec \beta$. (8)

(OR)

12. (a) Determine x and y given $x + iy = i^{\log(1+i)}$. (7)
- (b) Prove that the locus of all the points (x, y) satisfying $x + iy = \cos(u + iv)$ where u , and v are constants forms a circle with center at $(-1, 0)$ and radius $(\cosh v + \cos u)$. (8)

13. (a) Let i_1, i_2 , and i_3 be currents in a three-branch circuit. Kirchhoff's voltage law yields the simultaneous equations $i_1 + i_2 + i_3 = 6$; $i_1 + 2i_2 + 3i_3 = 10$; and $i_1 + 2i_2 + \alpha i_3 = \beta$. Use matrix method to find α, β for a unique set of currents in the branches. (7)

(b) Find the singular value decomposition of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$. (8)

(OR)

14. (a) Determine the matrix of rotation that removes the xy - term in the quadratic equation $3x^2 - 10xy + 3y^2 + 16\sqrt{2}x - 32 = 0$ and write the equation of the rotated conic. (7)

- (b) With the help of Cayley - Hamilton theorem, compute inverse of

the matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$. (8)

15. (a) A system whose closed - loop transfer function is

$$M(s) = \frac{K}{s(s^2 + s + 1)(s + 2) + K}. \text{ Determine the range of } K \text{ for}$$

which the system is stable. (7)

- (b) Using binomial expansion prove that

$$\frac{5}{3.6} + \frac{5.7}{3.6.9} + \frac{5.7.9}{3.6.9.12} + \dots \text{ to } \infty = \frac{-2}{3} + \sqrt{3}. \quad (8)$$

(OR)

16. (a) If a, b, c are the roots of the equation $y^3 - y - 1 = 0$ then calculate

$$\sum \left(\frac{1+a}{1-a} \right). \quad (7)$$

- (b) Determine all the roots of the algebraic equation, (8)

$8y^5 - 22y^4 - 55y^3 + 55y^2 + 22y - 8 = 0$ by the method of transformation.

17. (a) Find by double integration the area bounded by the parabola $y^2 = 4ax$ and its latus rectum. (7)

(b) The distance between two stations A and B located at $x = a$ and $x = b$ units respectively along a specific path is mathematically

modeled as $\int_a^b (x-a)^{m-1} (b-x)^{n-1} dx$. Use Beta and Gamma

functions to compute the integral when $a = 2, b = 4; n = m = 3$. (8)

(OR)

18. (a) A triangular prism is formed by planes whose equations are $2y = 3x, y = 0$, and $x = 2$. Determine the volume of the prism between the planes $z = 0$ and surface $z = 3 + xy$. (7)

(b) Determine the indefinite integral $\int_0^\infty \frac{dx}{\sqrt{1+x^4}}$ with the help of Beta and Gamma functions. (8)

PART - C

Answer the following

1 x 20 = 20 Marks

19. (a) The ellipsoid of inertia of a solid body is given by $\psi(X) \equiv 4x_1^2 + 4x_2^2 + x_3^2 - 2x_1x_2$. Obtain its canonical form in terms of a new orthogonal set of axes $\chi(y_1, y_2, y_3)$. Further, find the linear transformation that connects the two sets of coordinates and also discuss the nature of the quadratic form. (10)

(b) Change the order of integration in $I = \int_0^1 \int_{x^2}^{2-x} xy dx dy$ and hence evaluate the same. Sketch the region of integration. (10)

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